Tutorial 2_3

1. Illustrate at least five (5) disadvantages of a BJT that lead to the invention of a FET/MOSFET.
2. Explain with figures the effect of coupling, bypass and internal transistor capacitors on the frequency response behavior for a multistage amplifier.
3. What happened to an amplifier output waveform if you do not choose DC biasing properly?

Problem 1

a. Determine r_e.
b. Find Z_i (with $r_o = \infty\Omega$).
c. Calculate Z_o (with $r_o = \infty\Omega$).
d. Determine A_v (with $r_o = \infty\Omega$).
e. Repeat parts (c) and (d) including $r_o = 50\ \text{k}\Omega$ in all calculations and compare results.

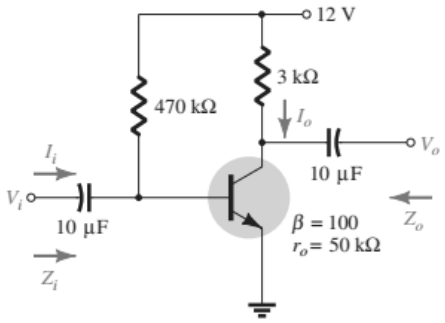


FIG. 5.25

Problem 2

a. r_e.
b. Z_i.
c. Z_o ($r_o = \infty\Omega$).
d. A_v ($r_o = \infty\Omega$).
e. The parameters of parts (b) through (d) if $r_o = 50\ \text{k}\Omega$ and compare results.

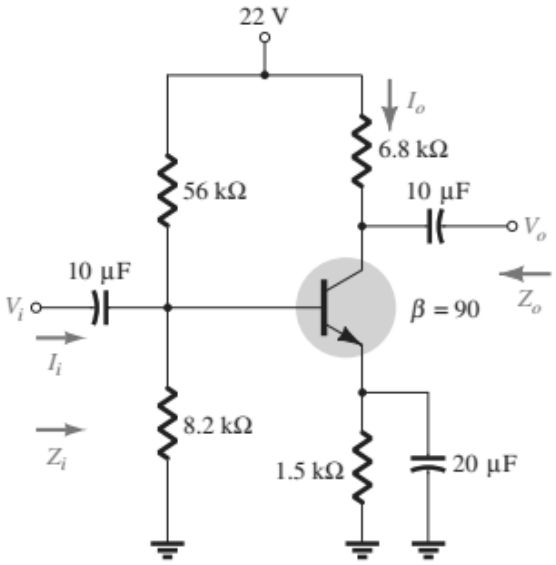
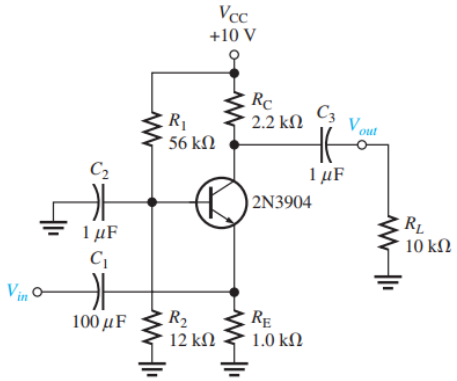


FIG. 5.28
Example 5.2.

Problem 3

Find the input resistance, voltage gain, current gain and power gain for the amplifier in Figure 1. β_{ac} = 250.



Problem 4

Determine the signal voltage at the base of the transistor if a 10 mv rms source with 300 ohm source resistance is applied in the input terminal. The emitter current Ie measured is 3.8 mA.

